

Literature Status: Sigma-Dedekind Complete Lattice Homomorphisms

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2026-06-23

Status. Literature already answered. The source question in arXiv:2007.14907 asks whether the theorem that every lattice homomorphism on an order-continuous Banach lattice attains its norm extends to σ -Dedekind complete Banach lattices. The later paper arXiv:2502.10165 explicitly answers this question negatively.

Source question. Dantas, Martinez-Cervantes, Rodriguez Abellan, and Rueda Zoca prove in arXiv:2007.14907, Section 4, that every lattice homomorphism on an order-continuous Banach lattice attains its norm. Immediately afterwards, around the paragraph containing their Example 4.2, they write that they do not know whether the theorem extends to σ -Dedekind complete Banach lattices.

Answering paper. Bilokopytov, Garcia-Sanchez, de Hevia, Martinez-Cervantes, and Tradacete, arXiv:2502.10165, Section 3, give equivalent lattice renormings of ℓ_∞ and $L_\infty[0, 1]$ with non-norm-attaining lattice homomorphisms. They explicitly state after Theorem 3.1 and its examples that these examples answer the question from arXiv:2007.14907 in the negative, because Dedekind completeness is an order property and is unchanged by equivalent lattice renorming.

Concrete counterexample. Let $X = \ell_\infty$ with its usual pointwise lattice order, but give it the equivalent lattice norm

$$\|x\| = \|x\|_\infty + \sum_{n=1}^{\infty} 2^{-n} |x_n|.$$

The underlying lattice is Dedekind complete, hence σ -Dedekind complete. Let $t \in \beta\mathbb{N} \setminus \mathbb{N}$, and let $\delta_t: C(\beta\mathbb{N}) \rightarrow \mathbb{R}$ be evaluation at t , identified with a free-ultrafilter limit on ℓ_∞ . This is a lattice homomorphism.

For every $x \in X$,

$$|\delta_t(x)| \leq \|x\|_\infty \leq \|x\|,$$

so $\|\delta_t\| \leq 1$ in the renormed space. If $u^{(m)} = \mathbf{1}_{\{n > m\}}$, then $\delta_t(u^{(m)}) = 1$ and

$$\|u^{(m)}\| = 1 + \sum_{n > m} 2^{-n} \longrightarrow 1,$$

so $\|\delta_t\| = 1$. If δ_t attained its norm at some x with $\|x\| \leq 1$, then $|\delta_t(x)| = 1$ would force $\|x\|_\infty = 1$ and $\sum_n 2^{-n} |x_n| = 0$. Hence $x = 0$, contradicting $|\delta_t(x)| = 1$. Thus δ_t does not attain its norm.

Scope limitation. This answers only the σ -Dedekind-complete extension question from the source paper. It does not settle Conjecture 5.5 of arXiv:2007.14907 about norm-attainment of $\widehat{x^*}$ on $FBL[E]$ versus norm-attainment of x^* on E .

References

- [1] S. Dantas, G. Martinez-Cervantes, J. D. Rodriguez Abellan, and A. Rueda Zoca, *Norm-attaining lattice homomorphisms*, arXiv:2007.14907.
- [2] E. Bilokopytov, E. Garcia-Sanchez, D. de Hevia, G. Martinez-Cervantes, and P. Tradacete, *Norm-attaining lattice homomorphisms and renormings of Banach lattices*, arXiv:2502.10165.